

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	30 September 2026
Team ID	SWTID-2026-4970
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Preparation	Prepare employee data in a spreadsheet and download it as .xlsx Create the custom table Employee Test (u_employee_test) Create the fields Employee ID, Employee Name, Email, Department and Location
FR-2	Data Import (Import Set)	Upload the spreadsheet through System Import Sets > Load Data Create the staging table Employee Import (u_employee_import) Load later files into the existing Import Set table
FR-3	Field Mapping & Transformation	Create the Transform Map "Sample Spreadsheet Import" (source: Employee Import, target: Employee Test) Map fields using Auto Map Matching Fields / Mapping Assist Run the Transform and confirm State Complete and Success
FR-4	Duplicate Prevention (Coalesce)	Enable Coalesce on the Employee ID field map Update existing records when the Employee ID matches Insert only new Employee IDs and ignore unchanged records
FR-5	Validation & Verification	View the imported records in the Employee Tests list Review Transform History (total, inserts, updates, ignored, skipped, errors) Check the Import Log for errors
FR-6	Reports	Employees by Department (Pie chart, Count) Employees by Location (Bar chart, Count) Employee List Report (List with all 5 columns)
FR-7	Dashboard & Sharing	Create the Employee Analytics Dashboard Add the three reports to the dashboard Share the dashboard with users, groups and roles

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The whole process uses ServiceNow's guided, menu-based screens (Load Data, Transform Map, Mapping Assist) and needs no coding. The target table has only five simple String fields and list columns can be personalised.
NFR-2	Security	Access is controlled by ServiceNow roles and ACLs; administrator access is needed to create tables, Transform Maps and the dashboard form (pa_dashboards ACL with Admin overrides). The dashboard can be restricted to selected users, groups and roles.
NFR-3	Reliability	Raw data is kept in a separate Import Set staging table, every load reports a State and Completion code, and Coalesce keeps the employee table consistent when the same data is imported more than once.
NFR-4	Performance	Records are processed in bulk instead of one by one: in the project run, 15 rows were processed in about 1.4 seconds and a 4-row update file in about 0.04 seconds.
NFR-5	Availability	The solution runs on the cloud-hosted ServiceNow instance and is used through a web browser, so no local installation is needed and reports and the dashboard are available whenever the instance is up.
NFR-6	Scalability	Import Sets and Transform Maps are built for bulk loads. The same Transform Map can be reused for new spreadsheets, and additional fields, tables or reports can be added without changing the process.